

Master's Business Analytics 2026-27

Professional vs Dual variant

Course Name	Code	EC	P1	P2	P3	P4	P5	P6
Compulsory for everyone		60 EC						
Mathematical Foundations of Machine Learning	XM_0189	6	6					
Applied Stochastic Modeling	X_400392	6	3	3				
Statistical Models	X_400418	6	3	3				
Research Seminar BA	XM_0054	6				3	3	
Master Project BA	X_400459	36						
Compulsory if Professional (Elective if Dual)		6 EC						
Project OBP	X_400213	6			6			
Compulsory if Dual		12 EC						
Dual Work period	XM_41010	12						

Specialization: Computational Intelligence								
Compulsory courses		18 EC						
Course Name	Code	EC	P1	P2	P3	P4	P5	P6
Evolutionary Computing	X_400111	6	6					
Deep Learning	XM_0083	6		6				
Natural Language Processing Technology	XM_0121	6					6	
at least 18 EC (12 EC if Dual or Professional with semester abroad)								
Compulsory selection								
Course Name	Code	EC	P1	P2	P3	P4	P5	P6
Data Mining Techniques	X_400108	6					6	
Dynamic Programming and Reinforcement Learning	XM_0093	6		6				
Large Language Models From Scratch*	XM_0188	6				6		
Learning Machines**	XM_0061	6			6			6
Machine Learning for the Quantified Self	XM_40012	6						6
Multi-agent Systems	XM_0052	6		6				
Performance of Networked Systems	X_405105	6		6				
<i>*machine learning (preferably, course Mathematical Foundations of Machine Learning) is a prerequisite</i>								
<i>**offered in p3 and p6, limited capacity</i>								

Specialization: Optimization of Business Processes								
Compulsory courses		18 EC						
Course Name	Code	EC	P1	P2	P3	P4	P5	P6
DPRL	XM_0093	6		6				
Mathematical Optimization	XM_0051	6		6				
Optimization of Business Processes	X_400422	6				3	3	
at least 18 EC (12 EC if Dual or Professional with semester abroad)								
Compulsory selection								
Course Name	Code	EC	P1	P2	P3	P4	P5	P6
Applied Analysis: Financial Mathematics (MSc)	XM_0140	6				3	3	
Continuous Optimization (LNMB)	XMM_400446	6	3	3				
Data Mining Techniques	X_400108	6					6	
Discrete Opt (LNMB)	XMM_400445		3	3				
Enterpreneurship in Analytics and AI		6				6		
Evolutionary Computing	X_400111	6	6					
Large Language Models From Scratch*	XM_0188					6		
Performance of Networked Systems	X_405105	6		6				
Scheduling Theory and Algorithms	XMM_0083	6				3	3	
Supply Chain Lab	E_TSCM_SCL	6				6		
<i>*machine learning (preferably, course Mathematical Foundations of Machine Learning) is a prerequisite</i>								

Specialization: Financial Risk Management								
Compulsory courses		18 EC						
Course Name	Code	EC	P1	P2	P3	P4	P5	P6
Stochastic Processes for Finance	X_400352	6	3	3				
Applied Analysis: Financial Mathematics (MSc)	XM_0140	6				3	3	
Investments for Business Analytics	XM_0187	6					6	
18 EC (12 EC if Dual or Professional with semester abroad)								
Compulsory selection								
Course Name	Code	EC	P1	P2	P3	P4	P5	P6
Advanced Asset Pricing*	E_FIN_AAP	6		6				
Advanced Financial Markets and Institutions	E_FIN_AFMI	6		6				
Data Mining Techniques	X_400213	6					6	
Derivatives**	E_FIN_DER	6					6	
Dynamic Programming and Reinforcement Learning	XM_0093	6		6				
Mathematical Optimization	XM_0051	6		6				
Quantitative Financial Risk Management**	E_FIN_QFRM	6					6	
Quantitative Investing*	E_FIN_QI	6	6					
<i>*out of Advanced Asset Pricing and Quantititave Investing, only one is allowed</i>								
<i>**course Investments is a compulsory prerequisite</i>								

Electives for everyone		up to 18EC (so 120 EC total)						
Course Name	Code	EC	P1	P2	P3	P4	P5	P6
Advanced Asset Pricing*	E_FIN_AAP	6		6				
Applied Analysis: Financial Mathematics (MSc)	XM_0140	6				3	3	
Applied Text Mining 1: Methods	L_PAMATLW004				6			
Continuous Optimization (LNMB)	XMM_400446	6	3	3				
Data Mining Techniques	X_400108						6	
Deep Learning	XM_0083	6		6				
Discrete Optimization (LNMB)	XMM_400445		3	3				
Enterpreneurship in Analytics and AI	XM_0090	6				6		
Advanced Financial Markets and Institutions	E_FIN_AFMI	6		6				
Investments for Business Analytics	XM_0187	6					6	
Large Language Models from scratch***	XM_0188					6		
Learning Machines****	XM_0061	6			6			6
Mathematical Optimization	XM_0051	6		6				
ML for the quantified self	XM_40012	6						6
Natural Language Processing	XM_0121	6					6	
Optimization of Business Processes	X_400422	6				3	3	
Performance of Networked Systems	X_405105			6				
Programming Large-Scale Parallel Systems	XM_40017	6	6					
Project OBP	X_400213	6						
Project Reinforcement Learning	XM_0120				6			
Quantitative Investing*	E_FIN_QI	6	6					
Queueing Theory	XMM_400397					3	3	
Scheduling Theory and Algorithms	XMM_0083	6				3	3	
Stochastic Processes for Finance	X_400352	6	3	3				
Supply Chain Lab	E_TSCM_SCL	6				6		
Time Series Models	E_EORM_TSM	6				6		

**out of Advanced Asset Pricing and Quantititave Investing, only one is allowed*

***course Investments is a compulsory prerequisite*

****machine learning (preferably, course Mathematical Foundations of Machine Learning) is a prerequisite*

*****offered in p3 and p6, limited capacity*

Recommendations on the order of courses:	
- If you wish to follow <i>Mathematical Optimization</i> and <i>Supply Chain Lab</i> , it is optimal to do <i>Supply Chain Lab</i> first and <i>Mathematical Optimization</i> next.	
- If you wish to follow <i>Mathematical Optimization</i> and <i>Discrete/Continuous Optimization</i> , it is optimal to do <i>Mathematical Optimization</i> first and <i>Discrete/Continuous Optimization</i> next.	
- If you wish to follow <i>Project Optimization of Business Processes</i> and the course <i>Optimization of Business Processes</i> , it is optimal to follow the project first and the course next.	
- If you wish to follow <i>Project Reinforcement Learning</i> , it is optimal to first do <i>Dynamic Programming and Reinforcement Learning</i> .	
- If you wish to follow <i>Large Language Models From Scratch</i> , it is optimal to first do <i>Mathematical Foundations of Machine Learning</i> .	
- It is optimal to follow <i>Deep Learning</i> in year 2.	